

- ▼ Foundational Topics
 - ▶ Prerequisites
 - ▶ VPC Fundamentals
 - ▶ Multiple VPCs
 - ▼ Security Controls
 - Network ACLs
 - Security Groups
 - Endpoint Policies
 - Clean Up
 - ▶ Connecting to On-Premises
 - ▶ Connecting to SDWAN
 - ▶ Network Monitoring
 - ▶ Network analytics using Amazon Quick Suite
- ▶ Advanced Topics
- Contributors & Revision History

Workshop catalog in AWS Builder Center [↗](#)

▼ Content preferences

Language

English ▼

Security Controls

ⓘ Please note, this module is optional and is not required to be completed to proceed with this workshop. You may skip to [Connecting to On-Premises](#)

ⓘ If you are running this lab at an **AWS-led event**, your facilitator has already set the region for you. If you are running it **self-service**, simply use the region you are working in. In either case, the region shown in the screenshots may not match your environment — this will not cause any problems.

If you are running this lab in **your own AWS Account**, it is recommended for all lab resources to be created in us-east-1 region so that the screenshots match your environment. This is not mandatory.

Prerequisites

- ▶ If you have not completed the Multiple VPCs section...

Previous

Next

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AWS Networking Workshop

workshop studio

Foundational Topics

Prerequisites

VPC Fundamentals

Multiple VPCs

Security Controls

Network ACLs

Security Groups

Endpoint Policies

Clean Up

Connecting to On-Premises

Connecting to SDWAN

Network Monitoring

Network analytics using Amazon Quick Suite

Advanced Topics

Contributors & Revision History

Workshop catalog in AWS Builder Center

Content preferences

Language

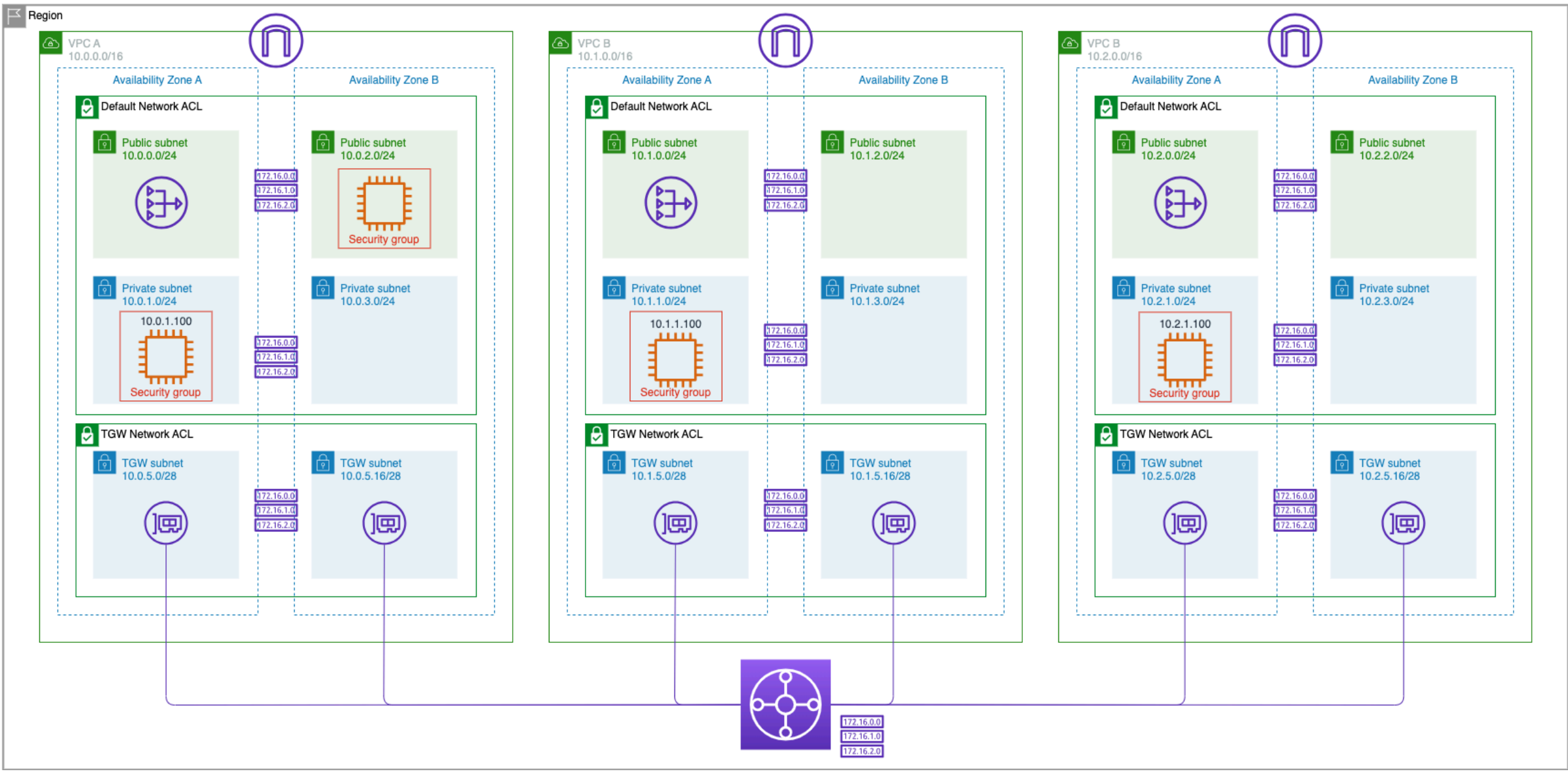
English

Network ACLs

Network ACLs are stateless access controls you configure at a subnet level, to allow or block a CIDR block on a particular port or range of ports. Network ACL rules are numbered list and evaluated top down, with a DENY ALL at the end. If a rule is matched, subsequent rules are not evaluated.

Both inbound and outbound traffic can be controlled with these rules. By default when you create subnets, they will be attached to the default Network ACL which has an ALLOW ALL rule for both inbound and outbound traffic.

In this section, we will modify the Network ACL associated with the workload subnets in VPC A to only ICMP traffic from VPC B's CIDR; and test connectivity from VPC A to VPC C, and test other connectivity from VPC B to VPC C as well.



Default NACL Configuration VPC A

- In the VPC Dashboard click on [Network ACLs](#)
- Select the check box for VPC A Workload Subnets NACL
- Click on the **Inbound Rules** tab below to view existing inbound rules

Network ACLs (1/6)

Filter network ACLs

Name	Network ACL ID	Associated with	Default	VPC ID	Inbound rules count	Outbound rules count
-	acl-0d41c57235546fac3	2 Subnets	Yes	vpc-0ac49c867a59c4e4 / VPC C	2 Inbound rules	2 Outbound rules
-	acl-0a56582596575ebdc	2 Subnets	Yes	vpc-0e1dca5e7188928b4 / VPC A	2 Inbound rules	2 Outbound rules
-	acl-0daae64ab10d35e5	2 Subnets	Yes	vpc-086a7d02f84ae4a22 / VPC B	2 Inbound rules	2 Outbound rules
☒ VPC A Workload Subnets NACL	acl-0a32d77d0f04a3afc	4 Subnets	No	vpc-0e1dca5e7188928b4 / VPC A	2 Inbound rules	2 Outbound rules
☐ VPC B Workload Subnets NACL	acl-02690ad2e010689...	4 Subnets	No	vpc-086a7d02f84ae4a22 / VPC B	2 Inbound rules	2 Outbound rules
☐ VPC C Workload Subnets NACL	acl-03b566c4784fb639c	4 Subnets	No	vpc-0ac49c867a59c4e4 / VPC C	2 Inbound rules	2 Outbound rules

acl-0a32d77d0f04a3afc / VPC A Workload Subnets NACL

Details

Inbound rules

Outbound rules

Subnet associations

Tags

You can now check network connectivity with Reachability Analyzer

Run Reachability Analyzer

Inbound rules (2)

Filter inbound rules

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

All traffic is evaluated for Protocol, Port and Source IP match. In this Network ACL, all traffic is allowed into the VPC A Public and Private Subnets by the first rule. The second rule which is a DENY ALL is not evaluated.

We will now modify the first rule (100) to allow only ICMP traffic from VPC B's CIDR.

- Click on **Edit inbound rules** button in the **Inbound rules** tab.
- For rule number 100 select ALL ICMP - IPv4 as Type and enter VPC B's CIDR of 10.1.0.0/16 for Source
- Click on **Save**

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the VPC.

Rule number

Type

Protocol

Port range

Source

Allow/Deny

100

All ICMP - IPv4

ICMP (1)

All

10.1.0.0/16

Allow

Remove

Add new rule

Sort by rule number

Preview changes

Save changes

- The screen should return to the Network ACL page and the updated rule will display on the **Inbound rules** tab like below
- Verify the rule for Type, Protocol, Port and Source for the 'ALLOW' rule 100.

Network ACLs (1/6)

Filter network ACLs

Name	Network ACL ID	Associated with	Default	VPC ID	Inbound rules count	Outbound rules count
-	acl-0c3edfee7f1ad9b4	2 Subnets	Yes	vpc-034b911a0e0d9d48d / VPC A	2 Inbound rules	2 Outbound rules
-	acl-09fb1e6cda03a61c	2 Subnets	Yes	vpc-07550fca322f3f35d / VPC B	2 Inbound rules	2 Outbound rules
☒ VPC A Workload Subnets NACL	acl-0321d1a316be0cfcf / VPC A Workload Subnets NACL	4 Subnets	No	vpc-07550fca322f3f35d / VPC B	2 Inbound rules	2 Outbound rules
☐ VPC B Workload Subnets NACL	acl-0f15051ee61e087df	4 Subnets	No	vpc-07550fca322f3f35d / VPC B	2 Inbound rules	2 Outbound rules
☐ VPC C Workload Subnets NACL	acl-09e6bdcedf3b798bf	4 Subnets	No	vpc-0988d5870c2e0db21 / VPC C	2 Inbound rules	2 Outbound rules

acl-0321d1a316be0cfcf / VPC A Workload Subnets NACL

Details

Inbound rules

Outbound rules

Subnet associations

Tags

You can now check network connectivity with Reachability Analyzer

Run Reachability Analyzer

Inbound rules (2)

Filter inbound rules

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	All ICMP - IPv4	ICMP (1)	All	10.1.0.0/16	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

We have now completed modifying the default Network ACL of VPC A to allow ONLY ICMP traffic from VPC B's CIDR and all other traffic will be denied by the catch-all DENY rule. Let us now test this from VPC B for ALLOW, and VPC C for DENY.

Note that we have not modified outbound rules, and the default outbound rule allows ALL traffic to flow out of the subnet.

Test Connectivity from VPC B to VPC A

Here, we will login to the EC2 instance in VPC B using EC2 using Session Manager, and verify reachability to the EC2 instance in VPC A over ICMP (ping)

- Click on [Instances](#)
- Select the check box for VPC B Private AZ1 Server and click **Connect** button on the top right

Instances (1/4)

Filter instances

Instance state: running

Clear filters

Connect

Instance state

Actions

Launch instances

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Private IP address
☐ VPC A Private AZ1 Server	i-079b	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a	10.0.1.100
☐ VPC A Public AZ2 Server	i-029b	Running	t2.micro	2/2 checks passed	No alarms	us-east-1b	10.0.2.100
☒ VPC B Private AZ1 Server	i-0b2i	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a	10.1.1.100
☐ VPC C Private AZ1 Server	i-0bf4	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a	10.2.1.100

- Click **Connect** in the **Session Manager** tab
- A new browser window opens with SSH connection to the instance in VPC B established and showing a command line prompt.
- Type this following command in the prompt:

```
ping 10.0.1.100 -c 5
```

```
sh-4.2$ ping 10.0.1.100 -c 5
PING 10.0.1.100 (10.0.1.100) 56(84) bytes of data.
64 bytes from 10.0.1.100: icmp_seq=1 ttl=254 time=1.39 ms
64 bytes from 10.0.1.100: icmp_seq=2 ttl=254 time=0.929 ms
64 bytes from 10.0.1.100: icmp_seq=3 ttl=254 time=1.00 ms
64 bytes from 10.0.1.100: icmp_seq=4 ttl=254 time=0.979 ms
64 bytes from 10.0.1.100: icmp_seq=5 ttl=254 time=0.943 ms
--- 10.0.1.100 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 0.929/1.050/1.399/0.180 ms
sh-4.2$
```

- The ICMP traffic should flow through and return as shown below.

We have now verified that the Network ACL on VPC A is allowing ICMP traffic to flow in and out from VPC B.

Test Connectivity from VPC C to VPC A

Now we will login to the EC2 instance in VPC C using EC2 using Session Manager, and check reachability to the EC2 instance in VPC A over ICMP (ping)

- Terminate the Session Manager connection to the EC2 instance in VPC B
- In the top left corner click [Instances](#)
- Select the check box for VPC C Private AZ1 Server and click **Connect** button on the top right

Instances (1/4)

Filter instances

Instance state: running

Clear filters

Connect

Instance state

Actions

Launch instances

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Private IP address
☐ VPC A Private AZ1 Server	i-079b	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a	10.0.1.100
☐ VPC A Public AZ2 Server	i-029b	Running	t2.micro	2/2 checks passed	No alarms	us-east-1b	10.0.2.100
☐ VPC B Private AZ1 Server	i-0b2i	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a	10.1.1.100
☒ VPC C Private AZ1 Server	i-0bf4	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a	10.2.1.100

- Click **Connect** in the **Session Manager** tab
- A new browser window opens with SSH connection to the instance in VPC C established and showing a command line prompt.
- Type this following command in the prompt:

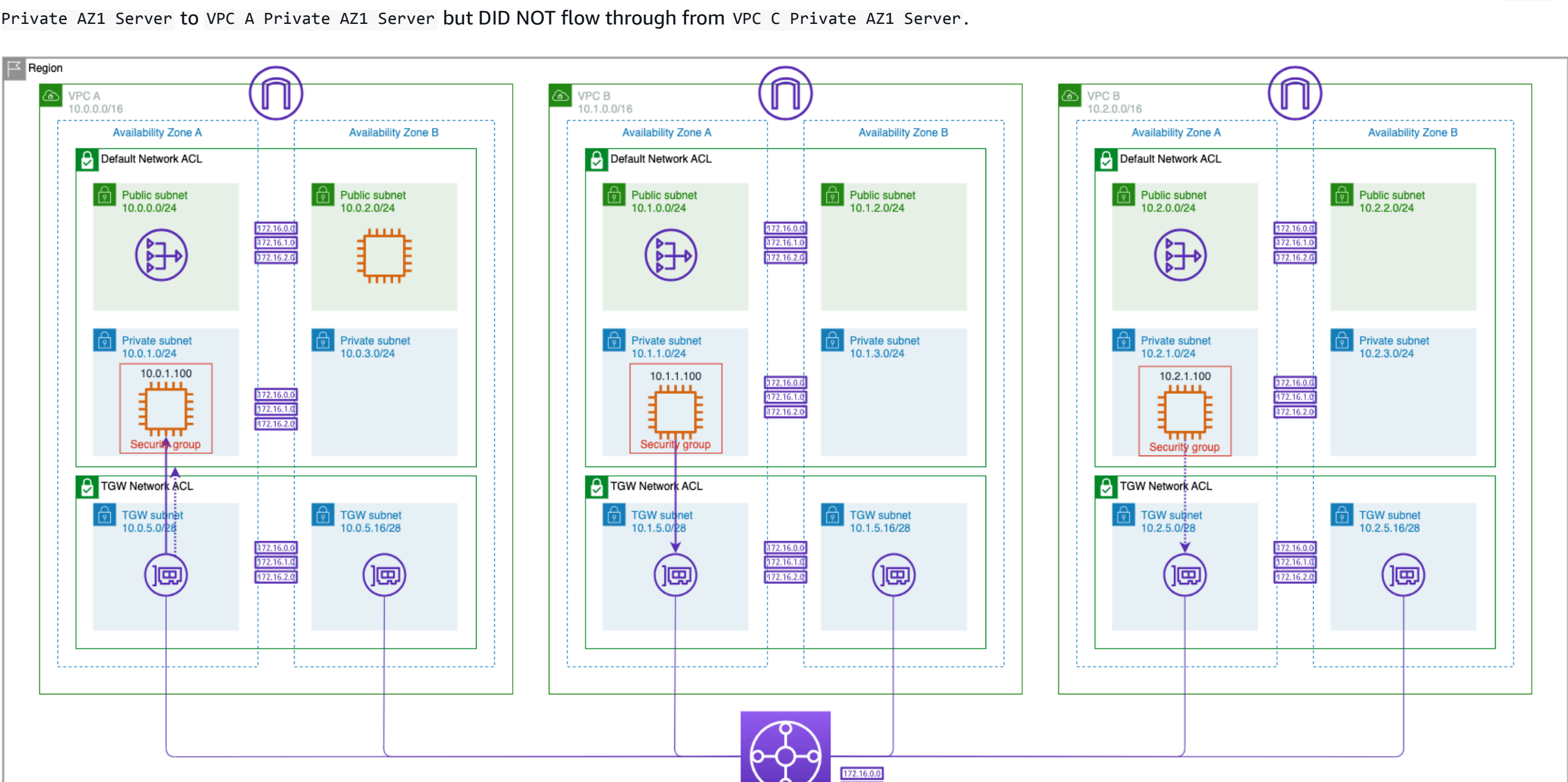
```
ping 10.0.1.100 -c 5
```

```
sh-4.2$ ping 10.0.1.100 -c 5
PING 10.0.1.100 (10.0.1.100) 56(84) bytes of data.
--- 10.0.1.100 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4072ms
sh-4.2$
```

The ping command will not generate a response because the Network ACL in VPC A is DENYING all ICMP traffic that does not originate from VPC B.

- Terminate the Session Manager connection

We modified the default Network ACL in VPC A to allow ICMP traffic only from VPC B; the only other rule is a DENY ALL. We verified that ICMP traffic flows through from VPC B Private AZ1 Server to VPC A Private AZ1 Server but DID NOT flow through from VPC C Private AZ1 Server.



Revert Changes

- In the VPC Dashboard click on [Network ACLs](#)
- Select the check box for VPC A Workload Subnets NACL

Network ACLs (1/6)

Filter network ACLs

Name	Network ACL ID	Associated with	Default	VPC ID	Inbound rules count	Outbound rules count
-	acl-0c3edfee7f1ad9b4	2 Subnets	Yes	vpc-034b911a0e0d9d48d / VPC A	2 Inbound rules	2 Outbound rules
-	acl-09fb1e6cda03a61c	2 Subnets	Yes	vpc-07550fca322f3f35d / VPC B	2 Inbound rules	2 Outbound rules
-	acl-0e2a81579f041786	2 Subnets	Yes	vpc-0988d5870c2e0db21 / VPC C	2 Inbound rules	2 Outbound rules
☒ VPC A Workload Subnets NACL	acl-0321d1a316be0cfcf	4 Subnets	No	vpc-07550fca322f3f35d / VPC B	2 Inbound rules	2 Outbound rules
☐ VPC B Workload Subnets NACL	acl-0f15051ee61e087df	4 Subnets	No	vpc-07550fca322f3f35d / VPC B	2 Inbound rules	2 Outbound rules
☐ VPC C Workload Subnets NACL	acl-09e6bdcedf3b798bf	4 Subnets	No	vpc-0988d5870c2e0db21 / VPC C	2 Inbound rules	2 Outbound rules

acl-0321d1a316be0cfcf / VPC A Workload Subnets NACL

Details

Inbound rules

Outbound rules

Subnet associations

Tags

You can now check network connectivity with Reachability Analyzer

Run Reachability Analyzer

Inbound rules (2)

Filter inbound rules

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

- Click on **Edit inbound rules** button in the **Inbound rules** tab
- Select Rule 100 and select ALL traffic for the Type enter a CIDR of 0.0.0.0/0 for Source
- Click on **Save**

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the VPC.

Rule number

Type

Protocol

Port range

Source

Allow/Deny

100

All traffic

All

All

0.0.0.0/0

Allow

Remove

Add new rule

Sort by rule number

Preview changes

Save changes

Network ACLs (1/6)

Filter network ACLs

Name	Network ACL ID	Associated with	Default	VPC ID	Inbound rules count	Outbound rules count
-	acl-0d41c57235546fac3	2 Subnets	Yes	vpc-0ac49c867a59c4e4 / VPC C	2 Inbound rules	2 Outbound rules
-	acl-0a56582596575ebdc	2 Subnets	Yes	vpc-0e1dca5e7188928b4 / VPC A	2 Inbound rules	2 Outbound rules
-	acl-0daae64ab10d35e5	2 Subnets	Yes	vpc-086a7d02f84ae4a22 / VPC B	2 Inbound rules	2 Outbound rules
☒ VPC A Workload Subnets NACL	acl-0a32d77d0f04a3afc	4 Subnets	No	vpc-0e1dca5e7188928b4 / VPC A	2 Inbound rules	2 Outbound rules
☐ VPC B Workload Subnets NACL	acl-02690ad2e010689...	4 Subnets	No	vpc-086a7d02f84ae4a22 / VPC B	2 Inbound rules	2 Outbound rules
☐ VPC C Workload Subnets NACL	acl-03b566c4784fb639c	4 Subnets	No	vpc-0ac49c867a59c4e4 / VPC C	2 Inbound rules	2 Outbound rules

acl-0a32d77d0f04a3afc / VPC A Workload Subnets NACL

Details

Inbound rules

Outbound rules

Subnet associations

Tags

You can now check network connectivity with Reachability Analyzer

Run Reachability Analyzer

Inbound rules (2)

Filter inbound rules

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

- The screen should return to the Network ACL page and the updated rule will display like below in the **Inbound rules** tab
- Verify the rule for Type, Protocol, Port and Source for the 'ALLOW' rule 100.

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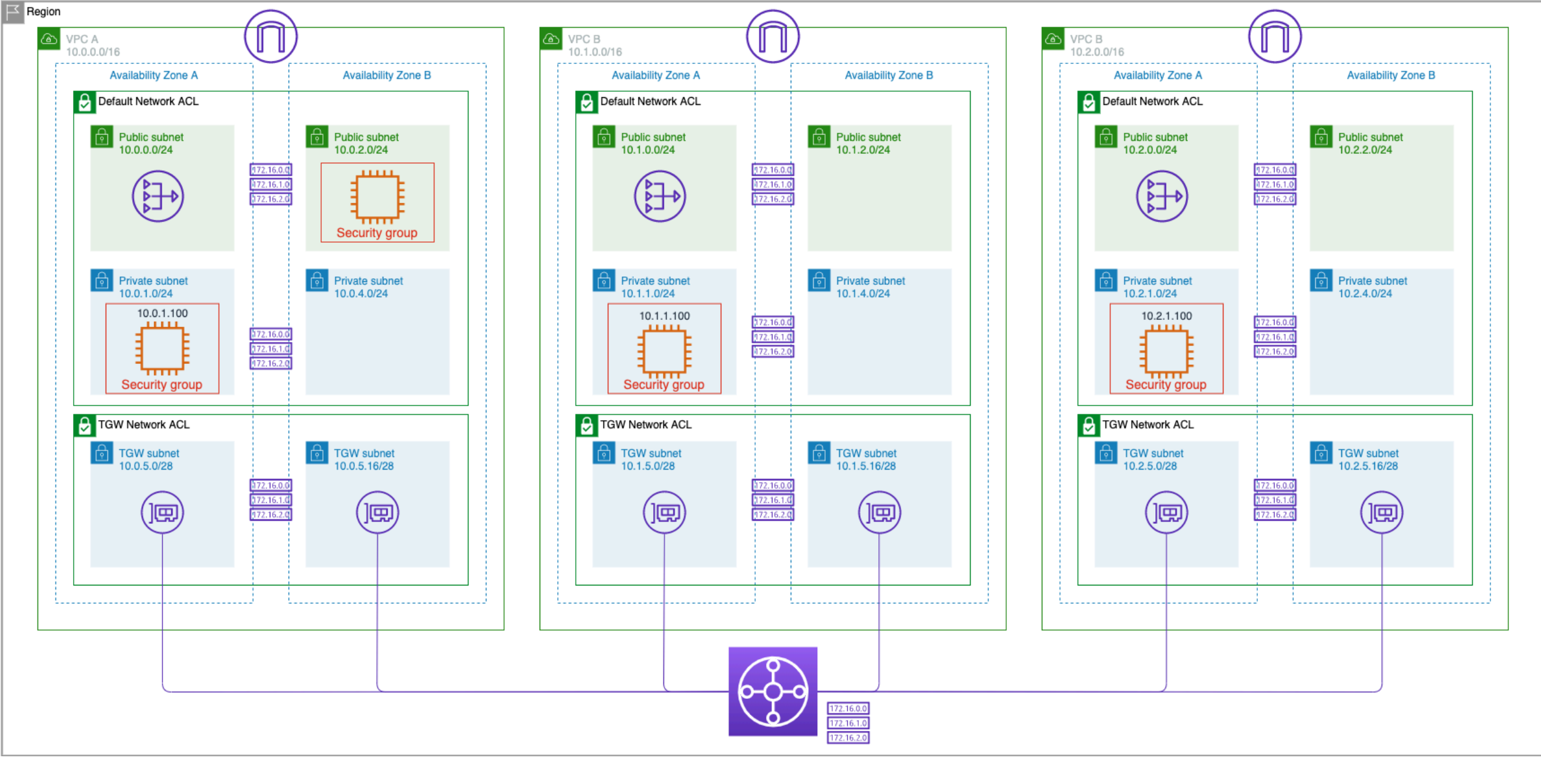
Decline

Customize

Security Groups

Security Groups are virtual, stateful firewalls attached to an instance or network interface. Both inbound and outbound rules can be defined to allow specific protocols, ports and source/destination CIDR. A DENY is not possible with security groups.

With Security Groups, all rules are evaluated before a network packet is allowed or blocked, unlike Network ACLs where the rules are evaluated in order of rule number and once a rule matches subsequent rules are not evaluated.



In this exercise, we will modify the security group attached to VPC A Private AZ1 Server to allow only ICMP traffic inbound from VPC C's CIDR only. We will verify that the EC2 instance in VPC C is able to ping the EC2 instance in VPC A, but that the EC2 instance in VPC B is not able to ping the same EC2 instance in VPC A.

Note: It is **NOT best practice** to have open Security Groups that allow everything (0.0.0.0/0). Limit access to what is required.

Modifying the Security Group in VPC A

- In the EC2 Dashboard click on [Instances](#)
- In the EC2 instance list, select the *check box* for VPC A Private AZ1 Server
- Scroll down to the **Security** tab and click on the Security Group link named something like sg-xxxxxxx (VPC A Security Group)

Instances (1/4) Info

Filter instances

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
☒	VPC A Private AZ1 Server	i-002f2b9aaba287f69	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a	-
☐	VPC B Private AZ1 Server	i-07b38ee58d411665c	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a	-
☐	VPC C Private AZ1 Server	i-07bdc9c955c282010	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a	-
☐	VPC A Public AZ2 Server	i-0176624421ee5b80c	Running	t2.micro	2/2 checks passed	No alarms	us-east-1b	-

Instance: i-002f2b9aaba287f69 (VPC A Private AZ1 Server)

Details Security Networking Storage Status checks Monitoring Tags

Security details

IAM Role: EC2ServerRole

Owner ID: 09

Launch time: Wed Sep 15 2021 15:06:47 GMT+1000 (Australian Eastern Standard Time)

Security groups: sg-034930cbda2e7fe4c (VPC A Security Group)

- In the Security Group page, click on the **Inbound rules** tab, and then click on **Edit inbound rules** button

sg-034930cbda2e7fe4c - VPC A Security Group

Details

Security group name: VPC A Security Group

Security group ID: sg-034930cbda2e7fe4c

Description: Open-up ports for ICMP and SSH

VPC ID: vpc-0e1dca5e7188928b4

Owner: 09

Inbound rules count: 2 Permission entries

Outbound rules count: 1 Permission entry

Inbound rules (2)

Filter security group rules

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
☐	-	sg-0560bc530a409a526	IPv4	All ICMP - IPv4	ICMP	All	0.0.0.0/0

Manage tags Edit inbound rules

- In the **Edit inbound rules** page, update the rule that is currently allowing ICMP from 10.0.0.0/8 to allow only from VPC C's CIDR 10.2.0.0/16

Inbound rules

Filter security group rules

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sg-0c	All ICMP - IPv4	ICMP	All	Custom	10.2.0.0/16

Add rule

- Click **Save rules**

Now we have modified the Security Group on the EC2 instance in VPC A allow ICMP traffic (ping traffic) only from sources in VPC C, and from nowhere else. We will now test to verify that we are NOT able to ping this instance from VPC B, and we are ABLE to ping from VPC C.

Test Connectivity from VPC B to VPC A

- In the EC2 Dashboard click on [Instances](#)
- Select the *check box* for VPC B Private AZ1 Server from the list of instances, and click **Connect** button

Instances (1/4) Info

Filter instances

Instance state: running

Clear filters

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Private IP address
☐	VPC A Private AZ1 Server	i-079i	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a	10.0.1.100
☐	VPC A Public AZ2 Server	i-029t	Running	t2.micro	2/2 checks passed	No alarms	us-east-1b	10.0.2.100
☒	VPC B Private AZ1 Server	i-0bz1	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a	10.1.1.100
☐	VPC C Private AZ1 Server	i-0rbf	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a	10.2.1.100

- In the 'Connect to instance' dialogue click **Connect** in the **Session Manager** tab. This will open up a session in a browser window.
- In the command line, try pinging the private IPv4 address of the EC2 instance in VPC A using the command

```
ping 10.0.1.100 -c 5
```

It will freeze and make no progress. This is because the Security Group on the EC2 instance in VPC A has inbound rules to only allow ICMP from VPC C. There are no rules for allowing VPC B, and an implicit DENY occurs.

```
sh-4.2$ ping 10.0.1.100 -c 5
PING 10.0.1.100 (10.0.1.100) 56(84) bytes of data.
--- 10.0.1.100 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4091ms
sh-4.2$
```

Now connect to the EC2 instance in VPC C and try to ping the EC2 instance in VPC A.

Test Connectivity from VPC C to VPC A

- Terminate the Session Manager connection and click [Instances](#)
- Select the *check box* for VPC C Private AZ1 Server from the list of instances, and click **Connect** button
- In the 'Connect to instance' dialog, click **Connect** in the **Session Manager** tab.
- In the command line, try pinging the private IPv4 address of the EC2 instance in VPC A using the command

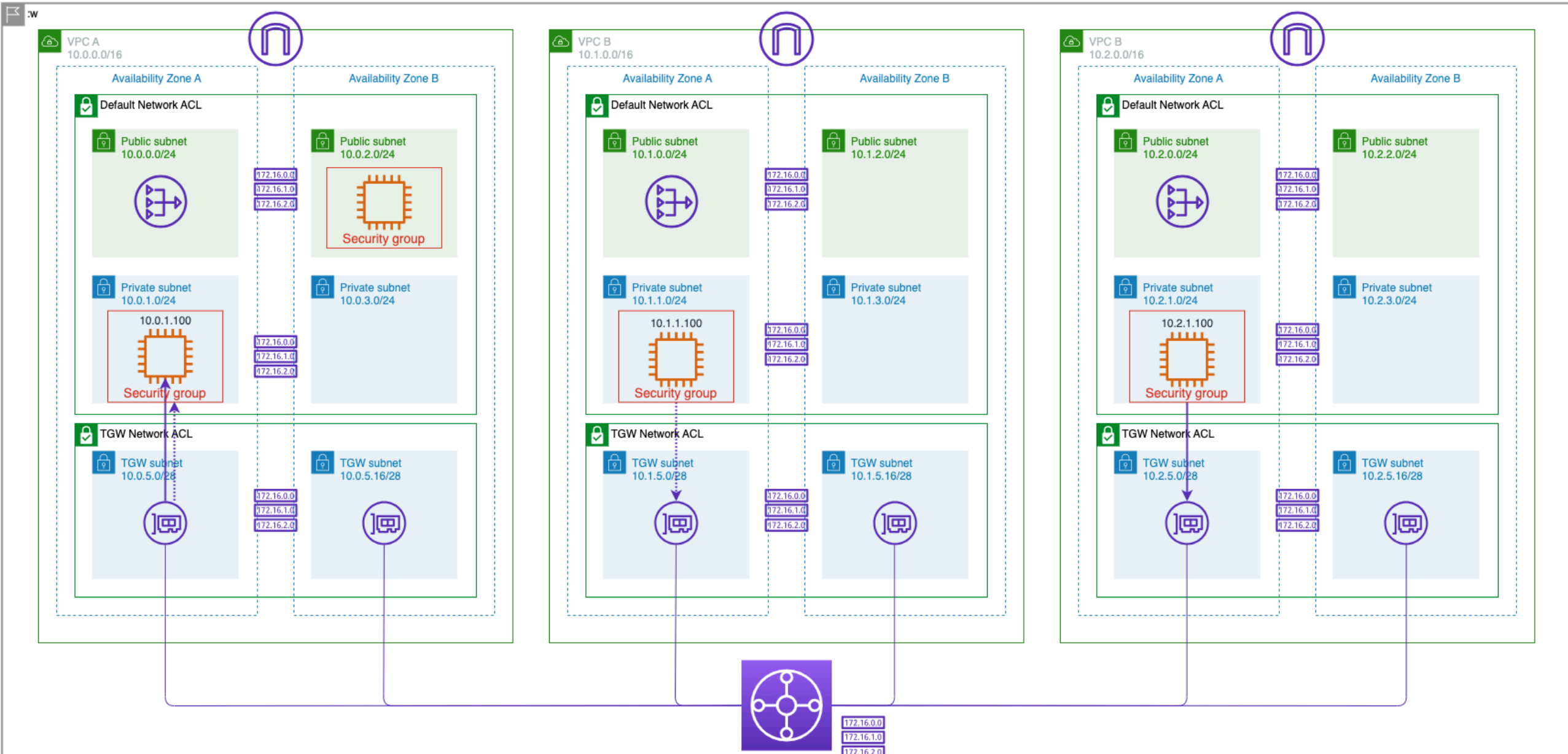
```
ping 10.0.1.100 -c 5
```

The ping will succeed and traffic will flow through. This is because the Security Group on the EC2 instance in VPC A is allowing ICMP traffic from VPC C's CIDR range.

```
sh-4.2$ ping 10.0.1.100 -c 5
PING 10.0.1.100 (10.0.1.100) 56(84) bytes of data.
64 bytes from 10.0.1.100: icmp_seq=1 ttl=254 time=0.998 ms
64 bytes from 10.0.1.100: icmp_seq=2 ttl=254 time=0.782 ms
64 bytes from 10.0.1.100: icmp_seq=3 ttl=254 time=0.813 ms
64 bytes from 10.0.1.100: icmp_seq=4 ttl=254 time=0.827 ms
64 bytes from 10.0.1.100: icmp_seq=5 ttl=254 time=0.720 ms
--- 10.0.1.100 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4019ms
rtt min/avg/max/mdev = 0.720/0.828/0.998/0.092 ms
sh-4.2$
```

- Terminate the Session Manager connection.

We have modified the Security Group on the EC2 instance in VPC A to allow only ICMP traffic from VPC C. We tested and verified that we cannot ping this instance from VPC B, but we are able to ping it from VPC C confirming the behavior of the Security Group.



Reverting Changes to the Security Group in VPC A

- In the EC2 Dashboard click on [Instances](#)
- In the EC2 instance list, select the *check box* for VPC A Private AZ1 Server
- Scroll down to the **Security** tab and click on the Security Group link named something like sg-xxxxxxx (VPC A Security Group)

Instances (1/4) Info

Filter instances

Connect Instance state Actions Launch instances

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
☒	VPC A Private AZ1 Server	i-002f2b9aaba287f69	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a	-
☐	VPC B Private AZ1 Server	i-07b38ee58d411665c	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a	-
☐	VPC C Private AZ1 Server	i-07bdc9c955c282010	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a	-
☐	VPC A Public AZ2 Server	i-0176624421ee5b80c	Running	t2.micro	2/2 checks passed	No alarms	us-east-1b	-

Instance: i-002f2b9aaba287f69 (VPC A Private AZ1 Server)

Details Security Networking Storage Status checks Monitoring Tags

Security details

IAM Role: EC2ServerRole

Owner ID: 09

Launch time: Wed Sep 15 2021 15:06:47 GMT+1000 (Australian Eastern Standard Time)

Security groups: sg-034930cbda2e7fe4c (VPC A Security Group)

- In the Security Group page, click on the **Inbound rules** tab, and then click on **Edit inbound rules** button

sg-0f7accad2d9965c95 - VPC A Security Group

Details

Security group name: VPC A Security Group

Security group ID: sg-0f7accad2d9965c95

Description: Open-up ports for ICMP and SSH

VPC ID: vpc-00d261038928d0acd

Owner: 7

Inbound rules count: 2 Permission entries

Outbound rules count: 1 Permission entry

Inbound rules (2)

Filter security group rules

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
☐	-	sg-0e6b6cc13b22ba3eb	IPv4	All ICMP - IPv4	ICMP	All	10.2.0.0/16

Manage tags Edit inbound rules

- In the **Edit inbound rules** page, update the rule that is currently allowing ICMP only from VPC C's CIDR of 10.2.0.0/16 back to 10.0.0.0/8

Inbound rules

Filter security group rules

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sg-0e6b6cc13b22ba3eb	All ICMP - IPv4	ICMP	All	Anywh...	0.0.0.0/8

Add rule

Cancel Preview changes Save rules

- Click **Save rules**

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AWS Networking Workshop

<

▼ Foundational Topics

▶ Prerequisites

▶ VPC Fundamentals

▶ Multiple VPCs

▼ Security Controls

Network ACLs

Security Groups

Endpoint Policies

Clean Up

▶ Connecting to On-Premises

▶ Connecting to SDWAN

▶ Network Monitoring

▶ Network analytics using Amazon Quick Suite

▶ Advanced Topics

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Workshop catalog in AWS Builder Center

▼ Content preferences

Language

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▼

Endpoint Policies

Endpoint policies are IAM policies attached to VPC endpoints to restrict/grant permission to the service's API calls. For example, with an S3 endpoint, a policy can be attached to limit only read access to one or more S3 buckets from the VPC.

Verify the permissions for the Gateway Endpoint

Now let us now verify access to S3 and what actions are permitted from VPC A Private AZ1 Server.

1. In the EC2 console navigate to [Instances](#)
2. Select the *check box* for instance VPC A Private AZ1 Server
3. Click **Connect** and use "Session Manager" to open a command prompt
4. Issue the command

```
aws s3 ls | grep networking-day
```

5. Copy the name of the bucket beginning `networking-day`
6. Issue the command `aws s3 ls s3://<your-bucket-name>` (replace with name of bucket you created) to check if we can list the contents of the bucket.
7. Check that the command was successful and no errors returned. The bucket is empty and hence it did not return any listing.
8. Let us try to create a test file and upload to the S3 bucket.

- Issue the command below to create a new file

```
sudo touch /tmp/test.txt
```

- Then attempt to copy to S3 by typing (replace with name of your bucket)

```
aws s3 cp /tmp/test.txt s3://<your-bucket-name>
```

9. You will see that the command succeeds and the file is uploaded to the S3 bucket.
10. Confirm the bucket contains the new file `aws s3 ls s3://<your-bucket-name>` (replace with name of bucket you created)

```
sh-4.2$ aws s3 ls s3://networking-day-us-east-1-2021-10-05_06:28:47_0_test.txt
sh-4.2$
```

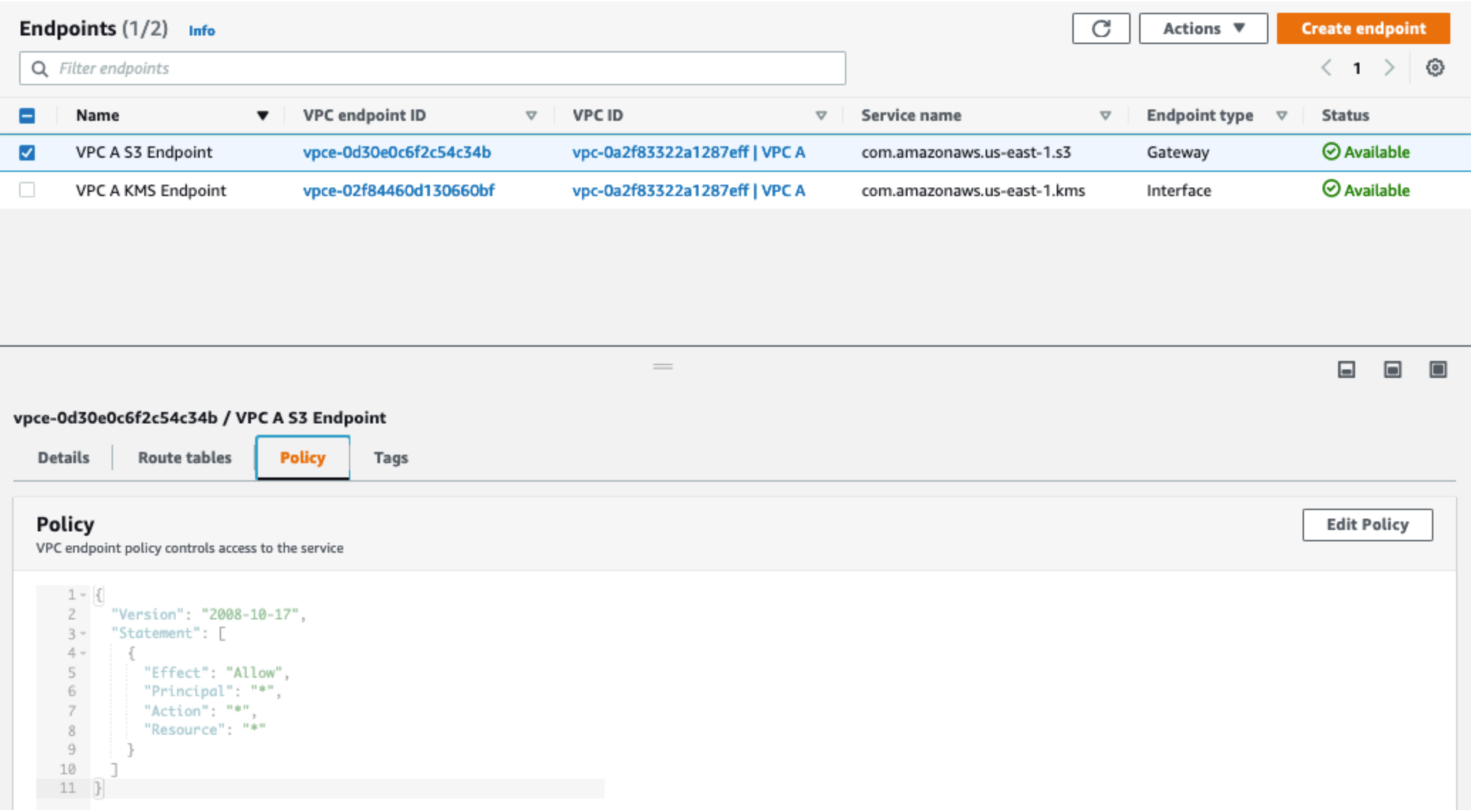
Update VPC Endpoint policy document to remove permissions

Let us now remove the permissions in the VPC Endpoint policy document that allow creating objects in the S3 bucket, and test it by attempting to upload a file to the bucket.

1. Navigate to VPC Dashboard and click on [Endpoints](#)
2. Select the Gateway Endpoint for S3 we created in the above steps.

If you did not begin this lab at VPC Fundamentals, your VPC Endpoints will not have Name tags set, unlike the screenshot below that shows the names. Select the Gateway Endpoint for S3 based on the Endpoint with the Service Name ending in `s3` and with the VPC ID ending in `VPC A`.

3. In the bottom frame, click on 'Policy' tab, then click on the Edit Policy button



4. In the policy editing screen,click on *custom*, enter the below policy document and click save. This policy removes `s3:Put*` permissions to the VPC Endpoint, effectively preventing instances from VPC A to upload objects to S3 buckets in the account.

```
{
  "Version": "2008-10-17",
  "Statement": [
    {
      "Sid": "ReadWriteAccess",
      "Effect": "Allow",
      "Principal": "*",
      "Action": [
        "s3:Get*",
        "s3:List*"
      ],
      "Resource": "*"
    }
  ]
}
```

5. Let us now try uploading the test file again by issuing the command `aws s3 cp /tmp/test.txt s3://<your-bucket-name>` (replace with name of bucket you created) to upload the file to the S3 bucket.

```
sh-4.2$ aws s3 cp /tmp/test.txt s3://networking-day-us-east-1-2021-10-05_06:28:47_0_test.txt
upload failed: ../../tmp/test.txt to s3://networking-day-us-east-1-2021-10-05_06:28:47_0_test.txt
An error occurred (AccessDenied) when calling the PutObject operation: Access Denied
sh-4.2$
```

Note that the command returned an 'Access Denied' error message when trying to upload a file to the S3 bucket. This is because the Endpoint policy allows only Get* and List* actions, effectively making the S3 bucket read only, and all other actions are denied.

Conclusion

We tested access to an S3 bucket through Gateway VPC Endpoint to S3, then we set the endpoint policy to allow only read actions; we tested that we can only list and get objects from the S3 bucket and verified that put objects are denied by the endpoint policy showing that the VPC Endpoint policy is an effective mechanism to control

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